

PLANAR RANDOM MOVEMENT AND THE BIVARIATE NORMAL DENSITY

J.M. van Zyl: Department of Mathematical Statistics, University of the Orange Free State, P.O. Box 339, Bloemfontein, 9300 Republic of South Africa.

Key words: Bivariate normal density; normal approximation; random movement.

Summary: A particle moves with discrete jumps on a plane. In this paper we derive the characteristic function of the distribution of the position of the particle after n jumps, consider the normal approximation, and derive approximate results concerning the distance of the particle from the origin after n jumps.

1. INTRODUCTION

A particle starting from the origin, jumps a random length U on the plane. The angle θ , with the horizontal axis is uniformly distributed on the interval $[0, 2\pi)$. In terms of polar co-ordinates, the distances travelled in the horizontal and vertical directions respectively for a jump, are $X=U \cos \theta$ and $Y=U \sin \theta$.

Let $\mathbf{x}_i' = (u_i \cos \theta_i, u_i \sin \theta_i)$, $i=1, \dots, n$, where x_i and x_j are independently distributed, $i \neq j$, denote n such displacements, and let $\mathbf{s}_n = \sum_{i=1}^n \mathbf{x}_i$. The co-ordinates of the position of the particle after n jumps are given by $\mathbf{s}_n$.

Let $r_n = (s'_n s_n)^{1/2}$ denote the direct distance from the origin after n movements. $Pr(R_n \leq r)$ is the probability that the particle will be within a circle of radius r after n jumps. In this paper we consider the relationship between the distribution of S_n and the bivariate normal density, and derive an approximate density for R_n .

These processes have a long history, and have been encountered in physics, astronomy and biology. There are many variations of the problem, for example the number of jumps can be a random variable, or the problem can be studied in more than two dimensions.

Pearson (1905) was interested in the density of R_n for the case in which the length of each jump is a fixed constant. Rayleigh (1905) mentioned that it followed from his results in the theory of sound that $f_{R_n}(r_n) = \frac{2}{n} r_n \exp(-r_n^2/n)$ is an approximate density for R_n when n is large and the jumps are of unit length. This was the origin of the Rayleigh density. Kluyver (1905) solved the problem for the case in which the distances are unequal known constants, $u_1, \dots, u_n$, and Rayleigh (1919) obtained the resulting density for R_n in three dimensions. It became known as the problem of random flights. A general result for p dimensions is given by Watson (1922, p. 421):

$$f_{R_n}(r_n; u_1, \dots, u_n | p) = \left[\Gamma\left(\frac{1}{2}p\right) \right]^{n-1} \cdot \left[r_n \int_0^\infty \left(\frac{1}{2}r_n t\right)^{\frac{1}{2}p-1} J_{\frac{1}{2}p}(r_n t) \prod_{m=1}^n \left[\frac{J_{\frac{1}{2}p-1}(u_m t)}{\left(\frac{1}{2}u_m t\right)^{\frac{1}{2}p-1}} \right] dt \right],$$

where J_γ denotes a Bessel function of the second type of order γ .

The random motion of particles in three dimensions attracted

much interest from physicists early in this century. In these processes particles move constantly. One way to pass from a jump process to such a process, is to consider the distribution of the particle after n movements, as the distribution of the increments of a particle that moves n times per unit of time.

By considering the process with small increments of time and displacement, it is possible to derive the density of the position of such a particle at time t , by using partial differential equations. Chandrasekhar (1943) gave a review of stochastic processes, including some results on random flights, and their applications in physics. Skellam (1951) gave solutions to random dispersal problems arising in biology and the theory of population growth, by using partial differential equations. Other results of interest are given by Whittle (1962), Paloheimo (1971), Kendall (1974), Mollison (1977), and in Feller (1971, pp. 29-33).

2. GENERAL RESULTS

Denote the density function of U by f_U , defined on some connected set $\mathcal{U} \subset \mathcal{R}^+$. Let μ_k^U denote the k -th moment of U about the origin. We assume that $\mu_2^U < \infty$. The problem can be formulated in a general way. Let a, b, α, μ_X and μ_Y be fixed real constants, $\mu' = (\mu_X, \mu_Y)$, $u' = (u a \cos \theta, u b \sin \theta)$, and let M be the orthogonal matrix

$$M = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}.$$

A general equation for the increments of a displacement is $x = Mu + \mu$. Let $\Sigma = M \Lambda M'$, where $\Lambda = \text{diag}(a^2, b^2)$. It is easy to see that $E(X) = \mu$ and $E((X-\mu)(X-\mu)') = (\mu_2^U/2)\Sigma$.

Let t_1, t_2 be real and $t' = (t_1, t_2)$. The characteristic function $\phi(t)$ of X is

$$(2.1) \quad \phi(t) = (1/2\pi) \int_{\mathcal{U}} f(u) \int_0^{2\pi} \exp(it'(\mu u + \mu)) d\theta du.$$

Let $p_1 = a(t_1 \cos \alpha - t_2 \sin \alpha)$ and $p_2 = b(t_1 \sin \alpha + t_2 \cos \alpha)$. Notice that $p_1^2 + p_2^2 = t' \Sigma t$.

$$(2.2) \quad \begin{aligned} \phi(t) &= \left[e^{it' \mu / 2\pi} \right] \int_{\mathcal{U}} f(u) \int_0^{2\pi} \exp(iup_1 \cos \theta + iup_2 \sin \theta) d\theta du \\ &= \left[e^{it' \mu / 2\pi} \right] \int_{\mathcal{U}} f(u) \int_0^{2\pi} \exp \left[iu(p_1^2 + p_2^2)^{1/2} \cos(\theta - \omega) \right] d\theta du, \\ &\qquad \qquad \qquad \omega = \tan^{-1}(p_2/p_1) \end{aligned}$$

$$= e^{it' \mu} \int_{\mathcal{U}} f(u) J_0(u(t' \Sigma t)^{1/2}) du$$

(Watson (1922, p 359)).

Assuming that the series made up of the terms $(-1)^k u^2 f(u) (t' \Sigma t)^k / 2^{2k} (k!)^2$ converges uniformly on $\mathcal{U}$ for some t_1, t_2 , it follows that

$$(2.3) \quad \phi(t) = e^{it' \mu} \sum_{k=0}^{\infty} (-1)^k (t' \Sigma t)^k \mu_{2k}^2 / 2^{2k} (k!)^2.$$

By letting $t_2 = 0$, $\alpha = 0$, $\mu_X = 0$, the k -th moment of X for k even can be derived as:

$$(2.4) \quad \begin{aligned} E(X^k) &= a^k \mu_k^2 / 2^k ((k/2)!)^2, \\ E(X^k) &= 0 \text{ for } k \text{ odd.} \end{aligned}$$

The characteristic function of S_n is $(\phi(t))^n$. An expression for $\phi(t)$ can also be found by changing the order of integration. Let Ψ denote the characteristic function of U . By an application of (2.2), it follows that

$$(2.5) \quad \phi(t) = \left[e^{it'\mu} / 2\pi \right] \int_0^{2\pi} \Psi((t'Zt)^{1/2} \cos(\theta - \omega)) d\theta, \\ \omega = \tan^{-1}(p_2/p_1).$$

3. THE NORMAL DENSITY AND APPROXIMATIONS

Let U have the density $f_U(u) = p^2 u \exp(-u^2 p^2 / 2)$, $u, p > 0$, i.e. a Weibull or Rayleigh density function. It then follows (Watson (1922, p. 394)) that $\phi(t) = \exp(it'\mu - \frac{1}{2}p^{-2}(t'Zt))$, the characteristic function of a bivariate normal with mean μ and covariance matrix $p^{-2}Z$. An easier way to obtain this result is by noting that if U has a Rayleigh density, $X = U \cos \theta$ and $Y = U \sin \theta$ are independently distributed normal random variables, both with mean zero. This can be shown by using the result of Box and Muller (1958), and by noting that if a random variable α is uniformly distributed on the interval $[0, 1]$, then $(-t \ln \alpha)^{1/2}$ has a Rayleigh density. Multiplication by $Z^{1/2}$ and adding μ will then lead to a similar result.

It follows in general by an application of the central limit theorem, that the distribution of S_n tends to a bivariate normal distribution with mean $n\mu$ and covariance matrix $(n\mu'_2/2)Z$ for n large. The accuracy of a normality approximation will depend

on the approximation of $\sum_{i=1}^n u_i$, since $s_n = M \left[\sum_{i=1}^n u_i \right] + n\mu$. The

assumption of normality can be investigated by a corollary to the Berry-Esseen inequality (Serfling (1980, p. 33)) if $\mu'_3 < \infty$. Since

$$E|X|^3 = a^3 4\mu'_3 / 3\pi, \quad \text{where } X = a U \cos \theta,$$

$$(3.1) \quad \sup_{-\infty < x < \infty} \left| \Pr \left(\sum_{i=1}^n X_i < xa \sqrt{n\mu'_2/2} \right) - \Phi(x) \right| \leq \gamma \frac{8\sqrt{2} \mu'_3}{3\pi(\mu'_2)^{3/2} \sqrt{n}}$$

where $\gamma = 0.7975$. The conditions are similar for $Y = bU \sin \theta$.

4. THE DISTRIBUTION OF R_n

The distance of the particle from the origin is $r_n = \sqrt{s_n' s_n}$, after n jumps. Notice that $s_n' s_n = s_n' (M M') s_n$. Let $s_n' M = (\xi_n, \eta_n)$, $M'\mu = m$ and $\frac{1}{2}\mu'_2 A = \text{diag}(\sigma_1^2, \sigma_2^2)^{1/2}$. The problem of finding the density of $R_n = (\xi_n^2 + \eta_n^2)^{1/2}$ reduces then to finding the density of the square root of the sum of the squares of two independent normal variates. Weil (1953) derived this density and further aspects of it are given by Blischke and Halpin (1966). It would also be possible to find the density of the distance from a given point d , if the particle has started from a point c , by letting $E(S_n) = n\mu + c - d$. We will assume that $c = d$. Under the assumption that S_n has an approximate bivariate normal density, (ξ_n, η_n) will be approximately normally distributed with mean nm and covariance matrix $\text{diag}(n\sigma_1^2, n\sigma_2^2)$.

$$\text{Let } F = \exp \left[-n(m_1^2/\sigma_1^2 + m_2^2/\sigma_2^2) \right],$$

$$A = \frac{1}{4}(\sigma_2^{-2} - \sigma_1^{-2}), \quad B = m_1/\sigma_1^2, \quad C = m_2/\sigma_2^2,$$

$$D^2 = B^2 + C^2, \quad E = \frac{1}{4}(\sigma_1^{-2} + \sigma_2^{-2}) \quad \text{and} \quad \tan \varphi = m_2 \sigma_1^2 / m_1 \sigma_2^2.$$

The density of R_n is

$$(4.1) \quad f_{R_n}(r_n) = (n\sigma_1\sigma_2)^{-1} r_n \exp(-r_n^2 E/n) \cdot \left[I_0(Ar_n^2/n) I_0(Dr_n) + 2 \sum_{j=1}^{\infty} I_j(Ar_n^2/n) I_{2j}(Dr_n) \cos(2j\varphi) \right],$$

where I_0 denotes a Bessel function of the first type. Special cases can be derived (Miller (1975, pp. 28-31)). The first case is where $a = b$, resulting in a density of the form

$$(4.2) \quad f_{R_n}(r_n) = (n\sigma_1^2)^{-1} r_n \exp \left[-(r_n^2 + \mu_1^2 + \mu_2^2) / 2n\sigma_1^2 \right] \cdot I_0(r_n(\mu_1^2 + \mu_2^2)^{1/2} / n\sigma_1^2)$$

and

$$E(R_n) = \frac{1}{2}(\pi n a^2 \mu_2^2)^{1/2} \exp \left[-(\mu_1^2 + \mu_2^2) / n a^2 \mu_2^2 \right] \cdot {}_1F_1 \left[3/2, 1; (\mu_1^2 + \mu_2^2) / n a^2 \mu_2^2 \right].$$

If $\mu = 0$:

$$(4.3) \quad f_{R_n}(r_n) = (n\sigma_1\sigma_2)^{-1} r_n \exp(-r_n^2 / 2n\sigma_1^2) \cdot {}_1F_1 \left[1/2, 1; r_n^2(\sigma_2^2 - \sigma_1^2) / 2n\sigma_1^2\sigma_2^2 \right]$$

and

$$E(R_n) = \frac{1}{2}(\pi n \mu_2^2)^{1/2} (a^2/b) {}_2F_1 \left[1/2, 3/2; 1; (b^2 - a^2)/b^2 \right].$$

The expected values can be calculated by an application of

results in Gradshteyn and Ryzik (1980, p. 716 and p. 851). The expected total distance travelled by the particle is $n\mu$.

5. A SIMULATION STUDY

A simulation study has been carried out for the case where $f_Y(u) = 1$, $0 \leq u \leq 1$, $a = b$, $n = 25$, $a = 1$ and $\mu = 0$. This process has been repeated 5000 times. The approximation can be improved if the density of R_n is truncated, since $R_n \leq 25$. The mean distance travelled is 2,5551 and the expected value of R_n found by applying the approximation is 2,5583. The results are given in table 5.1.

Table 5.1: Expected and observed values of R_n if $f_Y(u) = 1$

R_n	Observed	Expected
0,0 - 0,4403	109	114,9
- 0,8513	294	301,5
- 1,2622	455	453,7
- 1,6732	599	556,6
- 2,0842	577	604,3
- 2,4951	604	600,1
- 2,9061	559	553,9
- 3,3171	468	479,6
- 3,7280	387	391,9
- 4,1390	290	303,3
- 4,5500	250	223,0
- 4,9609	144	156,1
- 5,3719	117	104,1
- 5,7829	65	66,3
- 6,1938	32	40,3
- 6,6048	30	23,4
- 7,0158	12	13,0
- 7,4267	5	6,9
- 7,8377	3	3,5

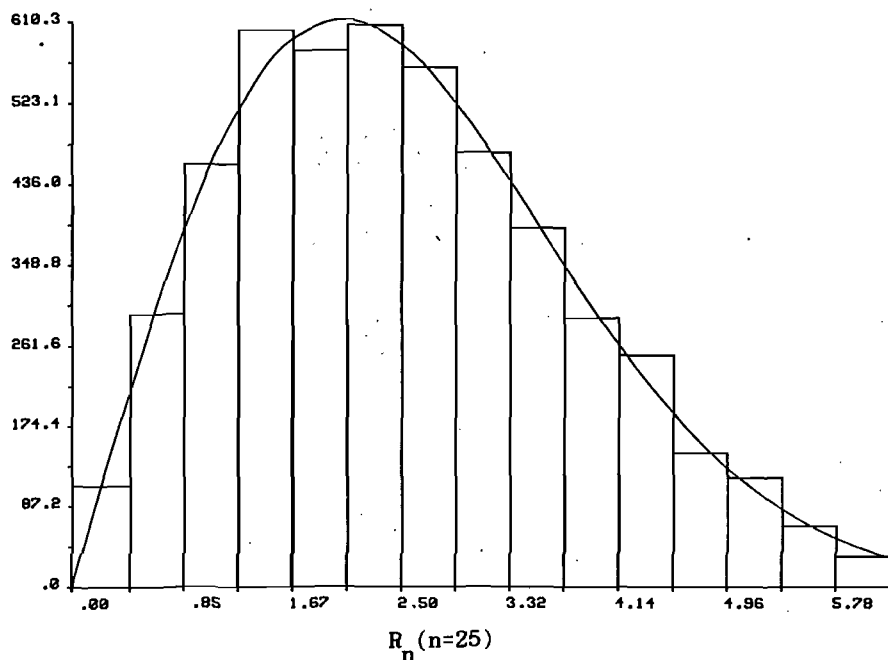


FIGURE 5.1: Expected and observed values of R_n

The simulation study was conducted on a Sperry 1170 computer at the University of the Orange Free State. Programs were written in FORTRAN level 11R1 and uniform random numbers were generated using the FORTRAN subroutine RANDU of the Sperry Math-Pack library.

REFERENCES

- BLISCHKE, W.R. & HALPIN, A.H. (1966). Asymptotic properties of some estimators of quantiles of circular error. *J.A.S.A.*, 61, 618-632.
- BOX, G.E.P. & MULLER, M.E. (1958). A note on the generation of random normal deviates. *Ann. Math. Statist.*, 29, 610-611.
- CHANDRASEKHAR, S. (1943). Stochastic problems in physics and astronomy. *Rev. Mod. Phys.*, 15, 1-89.
- FELLER, W. (1971). *An introduction to probability theory and its applications*, Vol. II. John Wiley, New York.
- GRADSHTEYN, I.S. & RYZIK, I.M. (eds.) (1980). *Tables of integrals, series and products, corrected and enlarged edition*. (Fourth edition). Academic Press, New York.
- KENDALL, D.G. (1974). Pole seeking Brownian motion and bird navigation. *J.R. Statist. Soc.*, B, 36, 365-402.
- KLUYVER, J.C. (1905). A local probability problem. *Proc. Section of Sci., K. Akad. van Wet. te Amsterdam*, VIII (1906), 341-350.
- MILLER, K.S. (1975). *Multivariate distributions*. Robert E. Krieger, New York.
- MOLLISON, D. (1977). Spatial contact models for ecological and epidemic spread. *J.R. Statist. Soc.*, B, 39, 283-326.
- PALOHEIMO, J.E. (1971). On a theory of search. *Biometrika*, 58, 61-75.
- PEARSON, K. (1905). The problem of the random walk. *Nature*, 77, 294.
- RAYLEIGH, LORD (1905). The problem of the random walk. *Nature*, 77, 318.
- RAYLEIGH, LORD (1919). On the problem of random vibrations, and of random flights in one, two or three dimensions. *Phil. Mag.*, 37, 321-347.

- SERFLING, R.J. (1980). *Approximation theorems of mathematical Statistics*. John Wiley, New York.
- SKELLAM, J.G. (1951). Random dispersal in theoretical populations. *Biometrika*, 38, 196-218.
- WATSON, G.N. (1922). *Theory of Bessel functions*. Cambridge University Press.
- WEIL, H. (1954). The distribution of radial error. *Ann. Math. Statist.*, 25, 168-170.
- WHITTLE, P. (1962). Topographic correlation, power-law covariance functions, and diffusion. *Biometrika*, 49, 305-314.